

HUMAN PERFORMANCE GROUP CHAIRMAN'S FACTUAL REPORT

ATTACHMENT C

MEETING NOTES, SEPTEMBER 27, 2006

(3 Pages)

September 27, 2006

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- Contract was C09B2
- The section of tunnel we call the “D Street Portal” is called the “C4A2” section by Modern Continental (MC).
- Contract specifications listed three (3) pre-approved adhesive anchor suppliers for Modern Continental to choose from.
- Selected Powers/Rawl from list.
- Product information for the adhesive and the installation procedure were submitted to B/PB and GF for review and approval.
- The contract called for 100% proof testing to 3250 lbs.
- After 5 anchors slipped in the HOV lanes, Modern Continental replaced the slipped anchors and tested to 6350 lbs.
- B/PB chose not to have MC test the “regular” eastbound and westbound lanes to the higher load.
- The installation procedure was as follows:
 1. Drill the holes using a core drill
 - Use of a core drill was specifically allowed in the contract and was in fact the preferred method after problems in the Ted Williams Tunnel.
 - The core drill was water-cooled.
 - The diamond bit was also specified in the contract.
 - The holes were 5” minimum and 5 1/2” maximum; the stop on the drill was used to make the hole the correct depth.
 - Holes were 5/8” diameter.
 2. Blow out the holes using compressed air
 - An electric compressor (oil-free?) was used
 3. Clean the hole with a wire brush.
 4. Blow the hole out again.
 - Hole depth was measured for every hole and recorded in a spreadsheet.
 5. Let the hole stand for 2-3 days.
 6. Insert the red cap
 7. Dispense and discard a small amount of adhesive
 8. Insert nozzle through cap
 9. Fill the hole at least halfway.
 10. Insert a pre-cut rod, twisting as it is inserted.
 - Rod was “bottomed out” in the hole
 11. Hold rod in place until adhesive sets up.
 - Length of rod visible/usable outside of hole measured.
 12. Adhesive is allowed to cure for 2-3 days.
 13. Rod/adhesive installation is tested (100% proof testing).

14. Plates attached.
 15. Rods hung and leveled with turnbuckles.
 16. Concrete panels placed in tracks.
 17. Concrete/tracks raised and attached to rods.
- An entire 5-panel module was raised and hung at one time.
 - MC noted that screen tubes should not be used in concrete applications.
 - Power-Fast installation guidelines specified in Rawl catalog, section 52.8.
 - On 7-29-99, a Newman or Powers representative was on-site to direct/assist/explain the install of 50 anchors in the NOV lanes, the “mock-up” area.
 - On 10-21-99, a Powers representative was on-site examining the HOV lane anchor failures.
 - The crew used to install the anchors was kept consistent and was comprised of experienced, core union employees.
 - Modern Continental has considerable experience with adhesive anchors, but believes the tunnel ceiling application is unique.
 - They are 99% sure the standard set version of the adhesive was used, but will check to be sure.
 - Once the tunnel was turned over to the MTA, they (MTA) had sole responsibility for maintenance and inspection.
 - MC has no documentation as to why there is no uni-strut in this section of the tunnel.
 - MC has been told by Powers that the current product is the same as the product sold previously and used in the tunnel.
 - MC believes 100% proof testing was required here due to earlier problems in the Ted Williams tunnel.
 - Although the contract required 100% proof testing, there was no provision for testing to failure.
 - Each module weighed 28,000 lbs.
 - MC will provide the NTSB with electronic copies of all materials provided to the AG.